RAJALAKSHMI ENGINEERING COLLEGE

RAJALAKSHMI NAGAR, THANDALAM – 602 105



CS23331 - DESIGN AND ANALYSIS OF ALGORITHMS

Laboratory Record Note Book

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College Roll No.:	251701039
Semester:	3rd Semester
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Exp.No	Name of the Experiment	Date of Completion
1	Coin Problem	31/08/2026
2	Cookies Problem	31/08/2026

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Pooja K 251701039-2025-CSD

P2

Started on	Monday, 31 August 2026, 3:26 PM
State	Finished
Completed on	Monday, 31 August 2026, 3:28 PM
Time taken	2 mins 13 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Write a program to take value V and we want to make change for V Rs, and we have infinite supply of each of the denominations in Indian currency, i.e., we have infinite supply of { 1, 2, 5, 10, 20, 50, 100, 500, 1000} valued coins/notes, what is the minimum number of coins and/or notes needed to make the change.

Input Format:

Take an integer from stdin.

Output Format:

print the integer which is change of the number.

Example Input :

64

Output:

4

Explanaton:

We need a 50 Rs note and a 10 Rs note and two 2 rupee coins.

Answer: (penalty regime: 0 %)

Ace editor not ready. Perhaps reload page?
Falling back to raw text area.

```
#include <stdio.h>

int main() {
    int V, count = 0;
    int denominations[] = {1000, 500, 100, 50, 20, 10, 5, 2, 1};
    int n = 9;

    scanf("%d", &V);

    for (int i = 0; i < n; i++) {
        count += V / denominations[i];
        V = V % denominations[i];
    }

    printf("%d", count);

    return 0;
}
```

	Input	Expected	Got	
✔	49	5	5	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Pooja K 251701039-2025-CSD

P2

Started on	Monday, 31 August 2026, 3:53 PM
State	Finished
Completed on	Monday, 31 August 2026, 3:55 PM
Time taken	1 min 37 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Assume you are an awesome parent and want to give your children some cookies. But, you should give each child at most one cookie.

Each child i has a greed factor g[i], which is the minimum size of a cookie that the child will be content with; and each cookie j has a size s[j]. If s[j] >= g[i], we can assign the cookie j to the child i, and the child i will be content. Your goal is to maximize the number of your content children and output the maximum number.

Example 1:

Input:

3

1 2 3

2

1 1

Output:

1

Explanation: You have 3 children and 2 cookies. The greed factors of 3 children are 1, 2, 3.

And even though you have 2 cookies, since their size is both 1, you could only make the child whose greed factor is 1 content.

You need to output 1.

Constraints:

1 <= g.length <= 3 * 10^4

0 <= s.length <= 3 * 10^4

1 <= g[i], s[j] <= 2^31 - 1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int compare(const void *a, const void *b) {
    return (*(int *)a > *(int *)b) - (*(int *)a < *(int *)b);
}

int main() {
    int n, m;


    scanf("%d", &n);

    int g[n];


    for (int i = 0; i < n; i++) {
        scanf("%d", &g[i]);
    }


    scanf("%d", &m);

    int s[m];


    for (int i = 0; i < m; i++) {
        scanf("%d", &s[i]);
    }


    qsort(g, n, sizeof(int), compare);
    qsort(s, m, sizeof(int), compare);

    int child = 0;
    int cookie = 0;

```

```
    while (child < n && cookie < m) {
        if (s[cookie] >= g[child]) {
            child++;
        }
        cookie++;
    }

    printf("%d", child);

    return 0;
}
```

	Input	Expected	Got	
✔	2 1 2 3 1 2 3	2	2	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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